

Who Converts Because of the Ad?

Uplift Targeting versus Response Targeting on 14 Million Criteo Users

Pablo Leyva
 R1 New Jersey Institute of Technology
pleyva2004@gmail.com

Abstract

Advertisers usually target the users most likely to convert. The causal question is different: which users convert *because of* the ad? We work through the full statistical-ML workflow on 13,979,592 users from Criteo’s randomized incrementality tests (v2.1), from exploratory analysis and hypothesis tests through predictive models, average-treatment-effect (ATE) estimators, heterogeneous-effect (CATE) and uplift learners, and a budget-constrained targeting optimizer. Three findings stand out. First, randomization is imperfect in the pooled benchmark: treatment is weakly predictable from the features (classifier two-sample AUC 0.509), and covariate-adjusted estimators agree with each other while sitting 13–28 % below the raw treated-versus-control difference (doubly robust visit effect +0.746 pp against a raw +1.034 pp). Second, assignment is not exposure: only 3.6 % of eligible users were shown an ad, and using assignment as an instrument the effect of actually seeing an ad is +21.2 pp on visits and +2.84 pp on conversions. Third, the central experiment has a split answer. For visits, uplift targeting at a 5 % budget finds 13,624 incremental visits against 8,542 for ranking by visit probability (+60 %); for conversions, no uplift learner beats ranking by conversion probability at any budget, because the ad lifts conversion roughly in proportion to baseline propensity. Targeting nevertheless pays: treating the top 10 % by conversion probability captures about 89 % of the lift of treating everyone. All estimates use a fixed train/validation/test protocol with bootstrap confidence intervals, and all code and results are released.

1 Introduction

An advertiser with a limited budget has to decide whom to show an ad to. The habitual answer is to rank users by their probability of converting and treat from the top. That ranking answers a *predictive* question: who is likely to buy? The decision the advertiser actually faces is *causal*: whose behaviour does the ad change? A user with a 40 % chance of buying anyway is a poor target if the ad moves that chance to 41 %, and a user with a 2 % chance who moves to 6 % is a good one. Ranking by response and ranking by effect coincide only if the ad’s lift is constant, or proportional to baseline propensity in a way that preserves the ordering.

This paper asks a narrow empirical question on a public benchmark: can we identify users who convert *because of* an ad, rather than users who would have converted anyway, and does targeting them beat targeting the likeliest converters?

We work on the Criteo Uplift Prediction dataset [5], 13,979,592 users pooled from several advertisers’ randomized incrementality tests, with a binary treatment (ad eligibility), two outcomes (site visit and conversion), and 12 anonymized features. Rather than fit a single uplift model, we run the whole classical statistical-ML workflow on one fixed train/validation/test protocol, so that every stage is checked against the stages before it.

Our findings, each with a confidence interval in Section 4:

- **Randomization is imperfect and it matters.** Treatment is weakly predictable from features, and the imbalance correlates with outcomes. Covariate-adjusted ATE estimators agree with each other and sit well below the raw difference.
- **Assignment is not exposure.** Only 3.6 % of eligible users were shown an ad. The naive exposed-versus-control contrast is confounded by user activity; an instrumental-variable estimate is not.
- **Uplift targeting wins for visits, not for conversions.** At a 5 % budget the X-learner finds about 60 % more incremental visits than a response model. For conversions, the response model is never beaten, and the causal forest is significantly worse on Qini.

Contributions. This paper makes three contributions:

1. An end-to-end, reproducible analysis of the Criteo v2.1 benchmark on the full 14M rows, with bootstrap intervals on every headline number and a test split that is touched once.
2. A careful treatment of the gap between randomized *assignment* and actual *exposure*, and of imbalance in the pooled benchmark, which together change the reported effect sizes by up to 28 %.
3. A direct, budget-matched comparison of uplift and response targeting that locates *where* uplift modeling helps (visits at small budgets, with at least 3M training rows) and where it does not (conversions).

Roadmap. Section 2 surveys related work. Section 3 describes the data, protocol, and estimators. Section 4 reports results. Section 5 interprets them and lists limitations, and Section 6 concludes with future work.

2 Related Work

Uplift and CATE benchmarks. The dataset we use is the v2.1 release of the Criteo benchmark of Diemert et al. [5], who advise modeling uplift on visits because conversions are too sparse, and who note that uplift models need very large samples. We test both claims directly.

Meta-learners and forests. The S-, T-, and X-learners are surveyed and compared in Künzel et al. [10]. The DR-learner follows Kennedy [9], and the transformed-outcome approach follows Athey and Imbens [1]. Honest causal forests are due to Wager and Athey [16]; we use the orthogonalized variant in EconML [2], built on the double machine-learning framework of Chernozhukov et al. [3].

Average-effect estimation. For ATE we use regression adjustment with full interactions and robust standard errors [11], inverse propensity weighting [15], and augmented inverse-propensity weighting (AIPW), which is doubly robust [14]. The effect of actually seeing an ad uses randomized assignment as an instrument [7].

Evaluating uplift. Because the individual effect is never observed, uplift rankings are scored with Qini and AUUC curves [13]. We complement them with the best-linear-predictor and sorted-group tests of Chernozhukov et al. [4], which ask whether estimated effects carry real heterogeneity and whether they are calibrated.

Randomization checks. To test whether treatment is predictable from covariates we use a classifier two-sample test [12].

Advertising measurement. Gordon et al. [6] compare observational approaches to randomized field experiments in advertising and find that adjustment often fails to recover the experimental answer. Our observational demonstration (Section 4.3) is in that spirit.

Gradient boosting. LightGBM [8] is our workhorse base learner for every predictive and meta-learner model.

3 Method

3.1 Data

Table 1 summarizes the Criteo Uplift v2.1 benchmark [5]. It merges several advertisers’ incrementality tests; in each, a random share of users is held out from ad targeting. Version 2.1 re-sampled every test to a common treatment ratio to remove a leak in v1. Rows were sub-sampled non-uniformly for privacy, so all rates describe the benchmark sample, not real-world ad incrementality.

Table 1: The Criteo Uplift v2.1 dataset as used here.

Item	Value
Users	13,979,592; split 60/20/20 into 8.39M train, 2.80M validation, 2.80M test, stratified on treatment $\times$ visit $\times$ conversion
Features	12 anonymized: 4 continuous, 8 hashed categoricals with 60–3,743 levels
treatment exposure	randomized ad eligibility; 85.0% treated shown ≥ 1 ad; 3.60% of treated users, 0 of control (post-randomization, self-selected)
visit / conversion	4.70% / 0.29% overall

3.2 Pipeline

Figure 1 shows the stages. Each reads a single configuration file (seeds, splits, sample sizes, model settings, business assumptions) and writes metrics, tables, and figures to disk; the narrative notebooks only read those outputs.

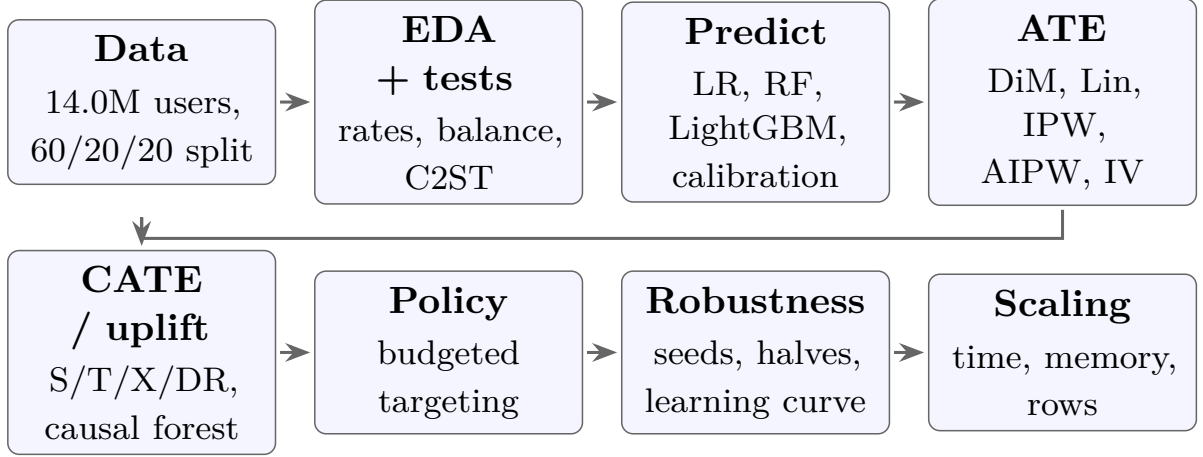


Figure 1: Analysis pipeline. Every stage uses the same stratified train/validation/test split.

3.3 Protocol

Features, calibrators, thresholds, hyperparameters, budget choices, and learner selection use only the train and validation splits. The test split is used once, for the reported numbers. Categorical levels seen fewer than 1,000 times in the training split are pooled into an “other” level, and the feature builder never sees a label.

Uncertainty comes from three sources: paired Poisson bootstraps over test users (uplift curves and policies), influence functions (ATE estimators), and refits with new seeds and disjoint training halves (robustness). Unless stated, intervals are 95%.

3.4 Predictive models

We fit $P(\text{conversion} \mid X)$ with L2 logistic regression (interpretable baseline), a random forest on a 2M-row sample, and LightGBM on all 8.39M training rows with early stopping on validation [8]. Model selection uses five-fold cross-validated log loss, a proper scoring rule that is far less noisy than PR-AUC with about 0.3% positives. Platt and isotonic calibrators are fit on validation and evaluated on test.

3.5 Average treatment effects

Write T for assignment, Y for an outcome, and X for features. The estimators are the difference in means, Lin regression adjustment [11] with HC2 standard errors, IPW (Horvitz–Thompson and Hájek) with design, logistic, and LightGBM propensities, and the doubly robust AIPW estimator [14],

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{T_i (Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - T_i) (Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} \right], \quad (1)$$

with cross-fitted LightGBM outcome models $\hat{\mu}_t$ and propensity $\hat{e}$; AIPW is our primary estimator. For the effect of actually seeing an ad, we use assignment T as an instrument for exposure D , with a Wald estimator and a covariate-adjusted variant [7]. It estimates the effect on the exposed under the exclusion restriction that assignment affects outcomes only through ads shown.

3.6 Heterogeneous effects and uplift

We implement S-, T-, X-, and DR-learners on LightGBM base learners, and fit an honest causal forest with EconML [2] on a 1M-row training sample. The transformed-outcome method is included as a class-variable baseline [1]. Every learner is compared with a *response model* $P(Y | X)$ that ignores treatment. Rankings are scored by the normalized Qini coefficient, the area between the Qini curve and random targeting divided by the area of a perfect ranking, with propensity-weighted bootstrap intervals. We also report the BLP and GATES diagnostics [4].

3.7 Targeting policies

Given a budget b (share of users treated), each policy treats the top b by its score: random, highest conversion probability, highest estimated uplift, and highest expected value, which treats every user with $\hat{\tau}(X)V - c > 0$ for value V and cost c . Policy value is estimated off-policy with propensity weights on the test split. For illustration, $V = \$50$ per conversion and $c = \$0.05$ per treated user; these are assumptions, not measurements.

4 Experiments

All numbers are read from `results/` and were produced by the full pipeline on the complete dataset. Percentage points are abbreviated pp.

4.1 Setup

Hardware is an Apple M4 Pro with 48 GB of memory and 12 threads. Seeds, splits, and sample sizes are fixed in a single configuration file. The development mode (a 5 % sample) runs the whole pipeline in minutes and is used only for debugging; it is not reported.

4.2 Classical tests and covariate balance

On all 14M users, treated users visit more (4.854 % versus 3.820 %, +1.034 pp [1.006, 1.063]) and convert more (0.309 % versus 0.194 %, +0.115 pp [0.108, 0.122]), a relative lift of +59.4 % [54.4, 64.7]. The null $p_T = p_C$ is rejected for both outcomes ($p < 10^{-170}$), and exact bootstrap intervals (2,000 draws) match the Wald ones. These are the *naïve* effects.

Under clean randomization, features should not predict treatment. They do, weakly: a classifier two-sample test has AUC 0.509 [0.508, 0.510] (permutation $p = 0.001$) and a joint likelihood-ratio test gives $p = 2 \times 10^{-53}$. The imbalance is correlated with the outcomes, which is why the next section adjusts for covariates.

4.3 Average treatment effects

Table 2 reports the ATE estimators. Every covariate-adjusted estimator sits well below the difference in means, and they agree with each other. The doubly robust estimate of the visit effect (+0.746 pp) is 28 % below the raw one; for conversion (+0.100 pp) the reduction is 13 %.

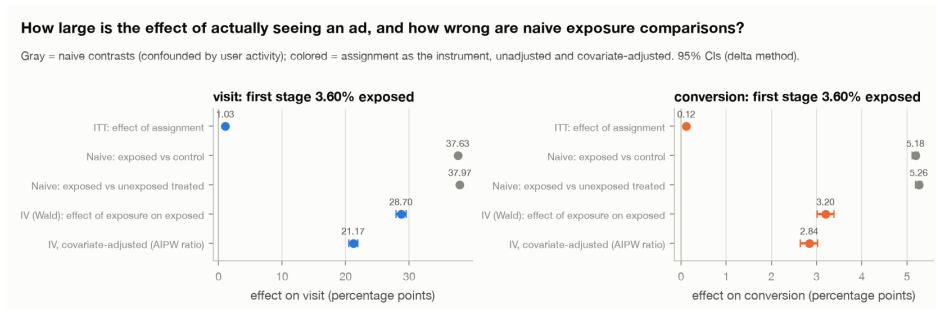
Table 2: ATE estimates in percentage points on all 14M users, with 95 % confidence intervals. g-computation has no interval reported.

Estimator	Visit	Conversion
Difference in means	1.034 [1.006, 1.063]	0.1152 [0.1085, 0.1219]
Lin regression adjustment	0.752 [0.726, 0.777]	0.0990 [0.0923, 0.1056]
IPW, Hájek, LightGBM propensity	0.761 [0.731, 0.791]	0.1007 [0.0935, 0.1079]
AIPW (doubly robust)	0.746 [0.721, 0.771]	0.1000 [0.0932, 0.1067]
g-computation (S-learner)	0.733	0.0977

Observational demonstration. To check that adjustment does what it claims, we make a confounded dataset out of the experiment: users are dropped with a probability that depends on a feature and on treatment, over 16 replicates. The naive difference then reads 2.19 pp for visits, with 0 % interval coverage of the full-data AIPW answer. AIPW recovers 0.749 pp (coverage 88 %) and Lin regression 0.748 pp.

4.4 Assignment versus exposure

Only 3.6 % of ad-eligible users were shown an ad. The naive exposed-versus-control contrast (+37.6 pp on visits) is confounded by user activity: active users are more likely to be exposed. Using randomized assignment as an instrument, the covariate-adjusted effect of actually seeing an ad is +21.2 pp [20.5, 21.9] on visits and +2.84 pp [2.65, 3.03] on conversions among exposed users.

**Figure 2:** Effect of actually seeing an ad, using assignment as an instrument, compared with naive exposure contrasts.

4.5 Predictive models

Table 3: Predictive models on the test split (2.8M users, base rate 0.29 %). Brackets are 95 % bootstrap intervals.

Model	ROC-AUC	PR-AUC	Log loss	Brier	Pred./base
Logistic regression	0.956 [0.953, 0.958]	0.214 [0.205, 0.224]	0.01202	0.00253	1.00
Random forest	0.953 [0.951, 0.956]	0.227 [0.217, 0.236]	0.01204	0.00250	0.99
LightGBM	0.958 [0.956, 0.960]	0.226 [0.217, 0.236]	0.01181	0.00251	1.00
LightGBM, scale_pos_weight=342	0.873	0.044	1.332	0.086	51.1

LightGBM beats logistic regression by a small but significant margin (paired bootstrap: ROC-AUC +0.0022 [0.0017, 0.0027], PR-AUC +0.012 [0.008, 0.016]) and ties the random forest on PR-AUC. The unweighted models are already calibrated: Platt and isotonic calibration change

log loss by at most 0.0001. Class weighting hurts ranking as well as calibration. Flagging the top 1 % of users catches 54 % of converters at 15.7 % precision, 54 times the base rate. Fourteen percent of test rows share a feature vector with a training row, but they hold only 32 of 8,155 test conversions, and excluding them leaves the metrics unchanged.

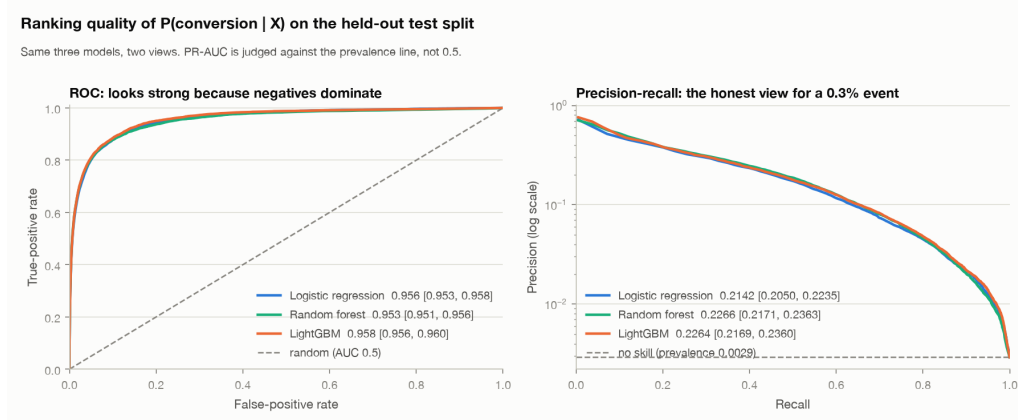


Figure 3: ROC and precision-recall curves. Precision-recall is the honest view of a 0.3 % event; ROC-AUC looks strong because negatives dominate.

4.6 Uplift models

Table 4 gives normalized Qini coefficients. On visits, the DR- and S-learners beat the response model (+0.0052 [0.0022, 0.0082] and +0.0048 [0.0011, 0.0082]). On conversions, every learner ties or trails it, and the causal forest is significantly worse (−0.0079 [−0.0132, −0.0030]).

Table 4: Normalized Qini coefficient on the test split, propensity-weighted, 95 % bootstrap intervals.

Scorer	Visit	Conversion
S-learner	0.0777 [0.0723, 0.0836]	0.1737 [0.1441, 0.2082]
T-learner	0.0713 [0.0659, 0.0773]	0.1339 [0.1040, 0.1640]
X-learner	0.0773 [0.0718, 0.0842]	0.1733 [0.1430, 0.2077]
DR-learner	0.0781 [0.0721, 0.0846]	0.1694 [0.1360, 0.2032]
Causal forest	0.0740 [0.0689, 0.0803]	0.1736 [0.1446, 0.2094]
Transformed outcome	0.0745 [0.0685, 0.0803]	0.1724 [0.1429, 0.2036]
Response model $P(Y X)$	0.0729 [0.0671, 0.0793]	0.1815 [0.1513, 0.2151]
Random	−0.0006	0.0048

The BLP heterogeneity test finds real heterogeneity for every learner ($p \leq 2 \times 10^{-8}$), and the S-learner’s visit slope is 1.02 [0.95, 1.09], meaning well-calibrated effect scores. Negative predictions are noise: users predicted to be *hurt* by the ad (23 % of users under the T-learner) have a positive observed effect (+0.13 pp [0.003, 0.26] on visits).

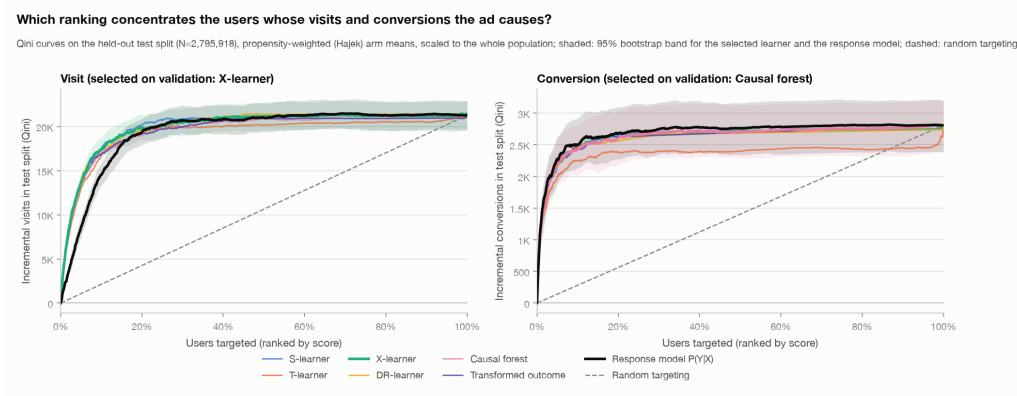


Figure 4: Qini curves on the 2.8M-user test split.

4.7 The central experiment: uplift versus response targeting

Table 5 gives incremental outcomes of uplift targeting minus response targeting at matched budgets. For visits, uplift targeting at a 5 % budget finds 13,624 [12,612, 14,555] incremental visits against 8,542 [7,418, 9,674] for the response model, a gain of +5,083 [3,960, 6,098], about 60 %. Of the visits in the response model’s top 5 %, 88 % would have happened without the ad; for the uplift model’s top 5 % the figure is 73 %. The advantage persists at 10 % and is gone by 20–30 %. For conversions, uplift is never significantly ahead.

Table 5: Incremental outcomes on the test split, **uplift minus response targeting** (95 % intervals). Visit uses the X-learner, conversion the causal forest.

Budget	Visit	Conversion
Top 5 %	+5,083 [3,960, 6,098]	−42 [−151, 65]
Top 10 %	+2,862 [1,953, 3,840]	−76 [−162, 24]
Top 20 %	+545 [−303, 1,335]	−71 [−127, 5]
Top 30 %	−258 [−1,040, 559]	−53 [−104, 9]

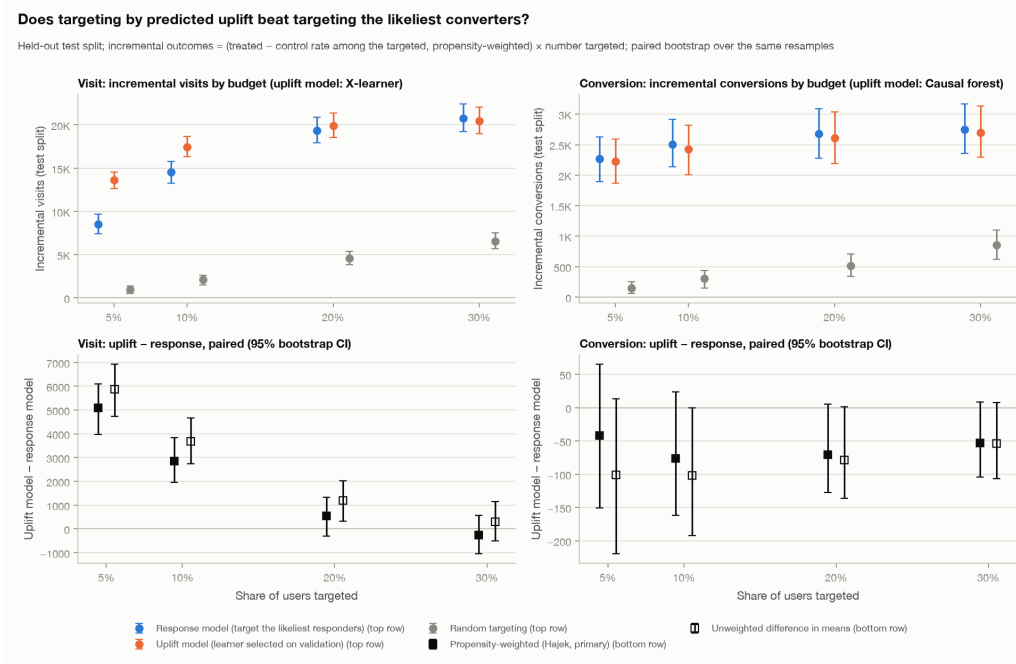


Figure 5: Uplift versus conversion-probability targeting, by budget.

The share of conversions that would have happened anyway is flat, about 67%, across score groups. The ad lifts conversion roughly in proportion to baseline propensity, so the likeliest converters are also among the most incremental.

4.8 Targeting policies

Table 6 compares policies on incremental conversions per 100K users. Treating the top 10% by conversion probability yields 89.5 [73.1, 104.3], against 100.4 [84.6, 114.7] for treating everyone: about 89% of the lift at a tenth of the treatment volume.

Table 6: Incremental conversions per 100K users, test split, propensity-weighted.

Policy	5 %	10 %	20 %	Treat all
Random	5.9 [2.2, 9.5]	12.3 [7.2, 16.8]	22.9 [16.1, 29.5]	100.4 [84.6, 114.7]
Highest conversion probability	81.2 [67.2, 95.1]	89.5 [73.1, 104.3]	95.9 [81.1, 110.0]	
Highest estimated uplift	79.7 [66.5, 93.2]	86.8 [71.5, 101.2]	93.3 [78.9, 107.2]	
Highest expected value	79.7	86.8	92.5 (treats 16.7%)	

With the budget chosen on validation, the economics favor targeting (Table 7). At a cost-to-value ratio of 0.001, blanket treatment nets about \$22 per 100K users, while response targeting at an 11% budget nets \$4,051 [3,330, 4,777].

Table 7: Net value per 100K users, budget chosen on validation.

Cost/value	Best policy	Budget	Net value	Treat all
0.0002	Response	31 %	\$4,628	\$4,022
0.0010	Response	11 %	\$4,051 [3,330, 4,777]	\$22 [−790, 758]
0.0020	Response	6 %	\$3,634	−\$4,978
0.0050	Response	3 %	\$2,775	−\$19,978

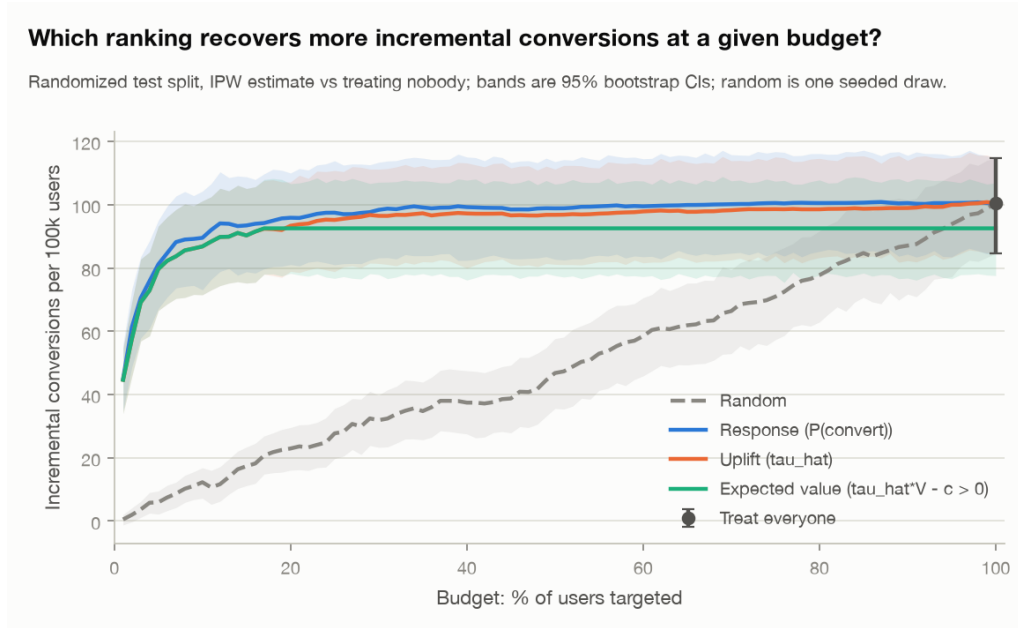


Figure 6: Incremental outcomes by treatment budget for each policy.

4.9 Robustness

Table 8 checks that the central result is not a seed or split artifact. The visit advantage is stable (+2,893, SD 144, always positive over three seeds) while the conversion deficit is small and consistently negative. Conversion effect scores are much less stable than visit scores (rank correlation between seeds 0.49 versus 0.98), as expected from a 0.3% event.

Table 8: Robustness of the uplift-versus-response comparison.

	Visit (X-learner)	Conv. (causal forest)
Uplift – response at 10%, over 3 seeds	+2,893 (SD 144), always > 0	–78 (SD 14), always < 0
$\hat{\tau}$ rank correlation between seeds	0.98	0.49
$\hat{\tau}$ rank correlation between disjoint halves	0.82	0.40
Top-10% set overlap between halves (Jaccard)	0.70	0.68

The X-learner’s visit Qini passes the response model’s only with at least 3M training rows (Figure 7), echoing the dataset authors’ finding that uplift needs very large samples. The pooled conversion model ranks equally well in each randomized arm (ROC-AUC 0.959 control, 0.958 treated) but, never having seen treatment, over-predicts conversion within control by 38% and under-predicts within the treated arm by 4%.

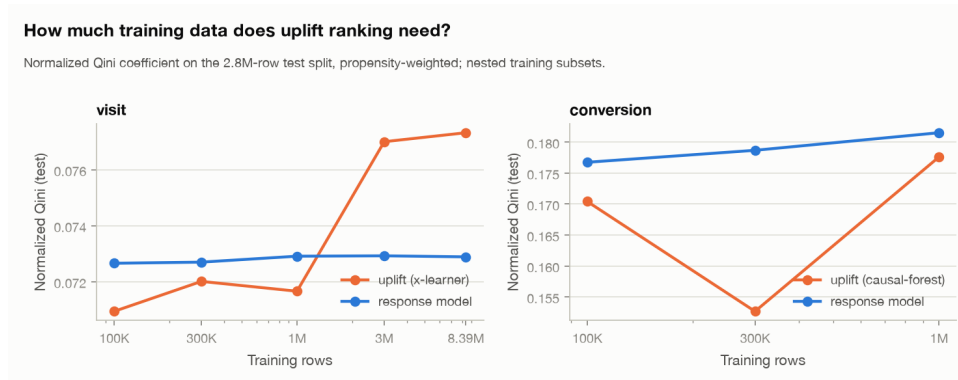


Figure 7: Visit learning curve: how much data uplift ranking needs before it overtakes the response model.

4.10 Scaling

On all 8.39M training rows, logistic regression fits in 7.5 s and peaks at 18.4 GB, while LightGBM (300 trees) fits in 18.5 s and peaks at 7.9 GB, with test PR-AUC 0.214 versus 0.224. LightGBM overtakes logistic regression on PR-AUC only above about 3M training rows. The full pipeline takes about 1.5–2 hours on this machine.

5 Discussion

5.1 Why visits and conversions disagree

The same learners, data, and protocol give opposite verdicts on the two outcomes. Two mechanisms plausibly combine, and we cannot cleanly separate them with this dataset.

1. **Signal.** Conversions are 0.29 % of users. The conversion effect scores are far less stable across seeds (rank correlation 0.49 versus 0.98), so an uplift ranking for conversions is mostly estimation noise layered on a signal that a simple response model already captures.
2. **Structure.** Independently of noise, the share of conversions that would have happened anyway is flat (about 67 %) across score groups. When lift scales with baseline propensity, the response ranking and the uplift ranking are nearly the same ranking, and there is nothing for uplift to add.

For visits the picture reverses: the highest-probability visitors are mostly “sure things” (88 % would have visited anyway), and a model that sees treatment can move budget to users the ad actually changes. This matches the dataset authors’ advice to model uplift on visits.

5.2 Randomization checks are not optional

The pooled benchmark is advertised as randomized, and the headline difference in means would be reported as such. A classifier two-sample test and a joint likelihood-ratio test show that treatment is weakly but detectably predictable, and that the imbalance correlates with the outcomes. Correcting it moves the visit effect by 28 %. The observational demonstration shows the adjusted estimators do recover the answer when a known confounder is injected, which supports, but does not prove, that the adjustment is doing the right thing on the real imbalance.

5.3 Assignment, exposure, and what “effect” means

The intent-to-treat effect (assignment) and the effect of seeing an ad differ by more than an order of magnitude on visits, because 96.4 % of eligible users are never shown an ad. A targeting policy acts on eligibility, so the ATE and uplift results above are the operationally relevant ones; the instrumental-variable estimate answers a different question, about users who do get exposed.

5.4 Limitations

The data is a benchmark, not a live system. Non-uniform sub-sampling, including negative sampling on labels, means absolute rates and effects are not real-world ad incrementality, and the business values in the targeting stage are assumptions.

Adjustment has limits. Treatment is only conditionally ignorable here, and adjusted estimators assume the 12 features capture the dependence between assignment and outcomes. If the sub-sampling depended on outcomes themselves, no feature adjustment fully corrects it. The mechanism behind the imbalance (per-test pooling or label-dependent sampling) is a hypothesis, not verified.

Anonymized features. Features are randomly projected, so heterogeneity findings describe feature segments, not interpretable user types.

Exposure estimates rest on the exclusion restriction. The IV estimates assume assignment affects outcomes only through ads shown.

Subsampled models. The random forest, logistic regression, and causal forest were trained on 1–2M-row samples for time. LightGBM and the meta-learners used all 8.39M training rows. The causal forest’s weaker conversion result may partly reflect this.

One dataset, one split. Every result comes from a single stratified split. Seed and disjoint-half refits address estimator variance but not variation across alternative splits or other advertisers’ data.

6 Conclusion

We worked through the classical statistical-ML workflow on 14M randomized advertising users and asked whether ranking by causal effect beats ranking by response. For visits, yes at small budgets: the X-learner finds about 60 % more incremental visits at a 5 % budget, provided it has at least 3M training rows. For conversions, no: no uplift learner beats ranking by conversion probability at any budget, because the ad lifts conversion in proportion to baseline propensity. Targeting still pays, since the top 10 % by conversion probability captures about 89 % of the lift of treating everyone at a tenth of the volume. Along the way, two corrections to naive analysis proved larger than any modeling choice: adjusting for imperfect randomization shrinks the visit effect by 28 %, and distinguishing assignment from exposure changes the estimand by an order of magnitude.

Future work. Several next steps are immediate.

1. **Locate the imbalance mechanism.** Test whether the treatment imbalance comes from per-test pooling or label-dependent sub-sampling, for example by recovering test identity from the categorical features and re-running the estimators within tests.
2. **Full-data causal forest.** Refit the causal forest and logistic regression on all 8.39M rows to check whether the conversion deficit of the forest is a sampling artifact.
3. **Composite targets.** Uplift on a value-weighted outcome (visit-then-convert) might recover signal that neither outcome has alone.
4. **Other splits and datasets.** Repeat the central comparison across resampled splits and on other public uplift benchmarks to test whether “uplift wins on the dense outcome, not the sparse one” holds beyond Criteo.

We release the code, configuration, test suite, and all full-data metrics, tables, and figures in the repository accompanying this paper.

References

- [1] Susan Athey and Guido W. Imbens. Machine learning methods for estimating heterogeneous causal effects. In *arXiv preprint arXiv:1504.01132*, 2015.
- [2] Keith Battocchi, Eleanor Dillon, Maggie Hei, Greg Lewis, Paul Oka, Miruna Oprescu, and Vasilis Syrgkanis. EconML: A Python package for ML-based heterogeneous treatment effects estimation, 2019. <https://github.com/py-why/EconML>.
- [3] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.
- [4] Victor Chernozhukov, Mert Demirer, Esther Duflo, and Iván Fernández-Val. Generic machine learning inference on heterogeneous treatment effects in randomized experiments. *arXiv preprint arXiv:1712.04802*, 2018.
- [5] Eustache Diemert, Artem Betlei, Christophe Renaudin, Massih-Reza Amini, Théophane Gregoir, and Thibaut Helbert. A large scale benchmark for individual treatment effect prediction and uplift modeling. *arXiv preprint arXiv:2111.10106*, 2021.
- [6] Brett R. Gordon, Florian Zettelmeyer, Neha Bhargava, and Dan Chapsky. A comparison of approaches to advertising measurement: Evidence from big field experiments at Facebook. In *Marketing Science*, 2019.
- [7] Guido W. Imbens and Joshua D. Angrist. Identification and estimation of local average treatment effects. *Econometrica*, 62(2):467–475, 1994.
- [8] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, 2017.
- [9] Edward H. Kennedy. Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2):3008–3049, 2023.
- [10] Sören R. Künnel, Jasjeet S. Sekhon, Peter J. Bickel, and Bin Yu. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165, 2019.

- [11] Winston Lin. Agnostic notes on regression adjustments to experimental data: Reexamining Freedman’s critique. *The Annals of Applied Statistics*, 7(1):295–318, 2013.
- [12] David Lopez-Paz and Maxime Oquab. Revisiting classifier two-sample tests. In *International Conference on Learning Representations*, 2017.
- [13] Nicholas J. Radcliffe. Using control groups to target on predicted lift: Building and assessing uplift models. In *Direct Marketing Analytics Journal*, 2007.
- [14] James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866, 1994.
- [15] Paul R. Rosenbaum and Donald B. Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983.
- [16] Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018.